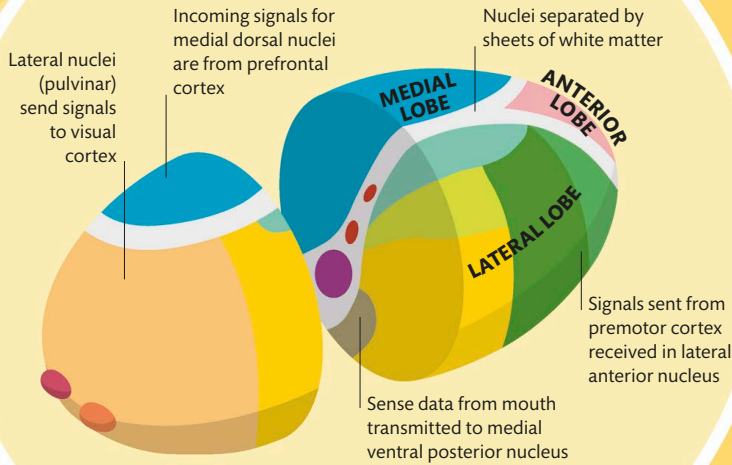




Thalamic nuclei

The thalamus is divided into three main lobes: the medial, lateral, and anterior. They are each further organized into zones, or nuclei, associated with particular sets of functions.

The thalamus

The word “thalamus” is derived from the Greek word for “inner chamber,” and this thumb-sized mass of gray matter sits in the middle of the brain, between the cerebral cortex and midbrain. It is formed from several bundles, or tracts, of nerves, which send and receive signals in both directions between the upper and lower regions of the brain, often in feedback loops (see p.91). It is associated with the control of sleep, alertness, and consciousness. Signals from every sensory system, except smell, are directed through the thalamus to the cortex for processing.

**WEIGHING JUST 0.1 OZ (4 G),
THE HYPOTHALAMUS IS
NOT MUCH LARGER THAN
THE END SEGMENT
OF A LITTLE FINGER**



The pituitary gland

Weighing about 0.01 oz (0.5 g), the tiny pituitary gland produces many of the body's most significant hormones under the direction of the hypothalamus. The hormones are released into the blood supply via a network of tiny capillaries. Pituitary hormones include those that control growth, urination, the menstrual cycle, childbirth, and skin tanning. Despite having the volume of a pea, the gland is divided into two main lobes, the anterior and posterior, plus a small intermediate lobe. Each lobe is devoted to the production of a particular set of hormones.

1 Stimulation

The hormones produced by the hypothalamus travel along axons to the pituitary gland.

2 Production

The chemicals from the hypothalamus stimulate the pituitary gland to release hormones.

3 Release

The netlike portal system collects the hormones and releases them into the bloodstream.

